

Project 3 Decision Trees, Bagging, and Overfitting (Report)

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Abstract—In this project I examine overfitting in Decision Tree Learners(DTlearner) Bagging Learners(Baglearner) and Random Tree Learners(RTlearner). By focusing on the effects of various leaf sizes on the root mean square error(RMSE), mean absolute error (MAE) and Max error to compare their performance. Bagging's influence on overfitting is also discussed. Quantitative comparisons on DTlearner and RTlearner are made using non-RMSE metrics. Model performance will be highlighted by supporting charts.

1 INTRODUCTION

In this paper I will investigate the behavior of different algorithms when it comes to learning in terms of their susceptibility to overfitting and impact on overall performance. The focus of the three experiments

1. DTlearner: A decision tree learner is tested across various leaf sizes to observe the effects of overfitting
2. Baglearner: Bagging is applied to examine whether it mitigates overfitting.
3. A comparison between DTlearner and RTlearner: this will be a quantitative comparison of decision trees and random trees using various metrics.

I hypothesized that overfitting would occur in DTLearner as model leaf size decreases and that bagging should reduce if not eliminate overfitting. DTlearner and RTlearner might have unique strengths depending on the situation.

2 METHODS

The experiment was carried out using the Istanbul.CSV dataset. The dataset was split into 60% percent training and 40% testing sets. Each experiment was then repeated for a variety of leaf sizes ranging from 1 to 50 in increments of 5.

Experiment 1 (DTlearner): The DTlearner model was trained and tested with different leaf sizes and RMSE was then used to evaluate performance.

Experiment 2 (Baglearner): Bagging was set to 20 bags and was applied to DTlearner and RMSE was again calculated for each leaf size to examine the effect that bagging would have on overfitting.

Experiment 3 Both learners were compared using three metrics: RMSE, Mean Absolute Error (MAE), and Max Error.

3 DISCUSSION

4 EXPERIMENT 1

I will look at Overfitting in DTlearner; overfitting is a phenomenon where a model performs well during training but very poorly on actual testing set.

Does overfitting occur?

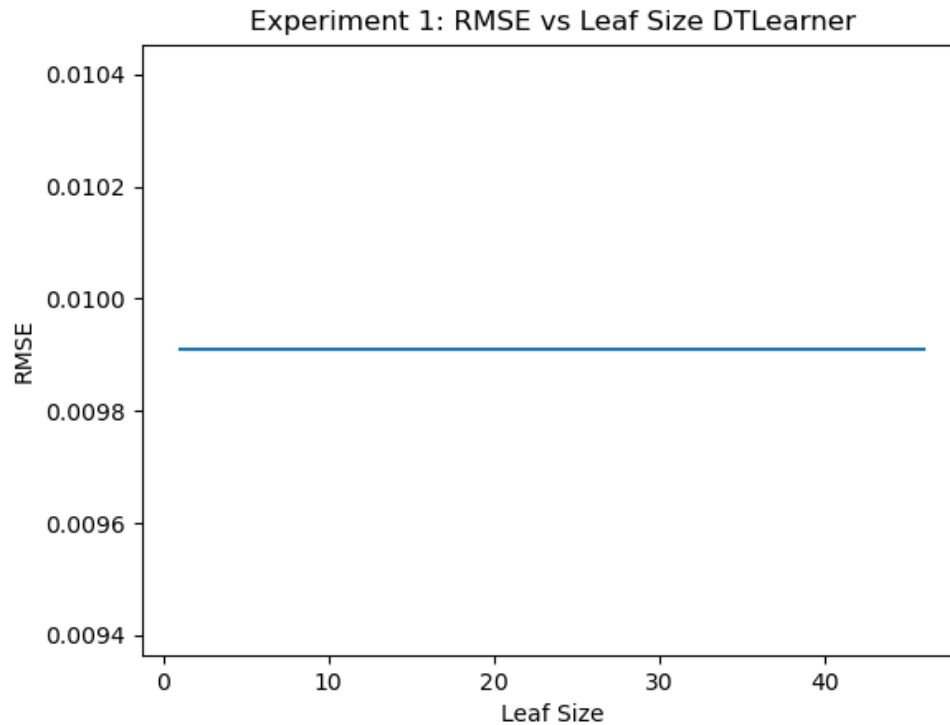
No overfitting was observed. The RMSE remained stable across all leaf sizes, suggesting that DTlearner maintained a similar performance no matter the model complexity.

For which leaf sizes does overfitting occur?

Since no overfitting was observed, there is no specific leaf size where overfitting occurs.

Direction and starting point of overfitting

Since no overfitting was observed, no starting point can be identified.



5 EXPERIMENT 2: BAGGING AND OVERFITTING

Bagging is a method that can reduce the amount of variance and overfitting by averaging predications of several learners. For experiment 2 I tested bagging to try to mitigate the overfitting in DTlearner.

Can bagging reduce overfitting?

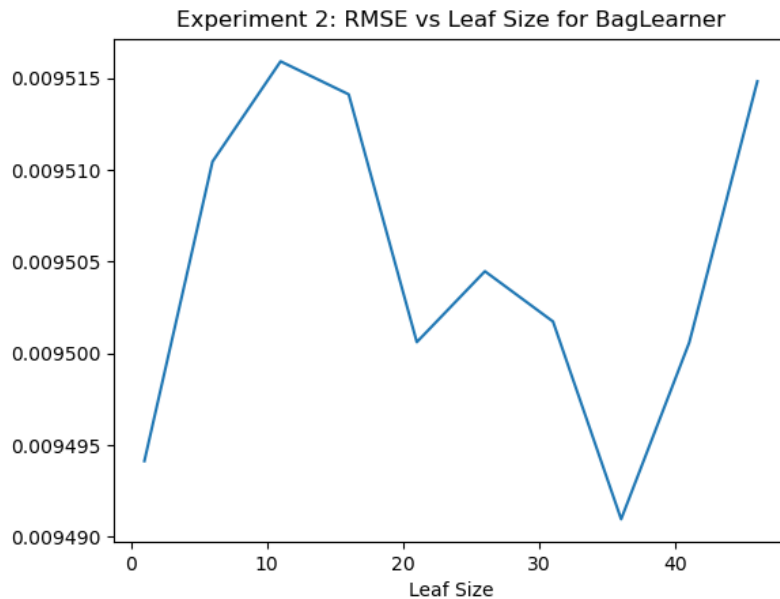
Yes bagging does reduce the seen effect of overfitting as seen as the RMSE stabilizes across various leaf sizes when compared to DTlearner that doesn't have bagging.

Can bagging eliminate overfitting?

Overfitting will still remain even with bagging so no however the severity is reduced.

Direction and starting point of overfitting

Minor increases in RMSE were observed starting around leaf size 40 however not severe enough to be considered overfitting.



EXPERIMENT 3 COMPARISON OF DTLEARNER AND RTLEARNER

For the final experiment I compared which was better between DTLearner and RTLearner using RMSE. RTLearner outperformed DTLearner with a lower RMSE, which shows RTLearner had better overall performance in terms of minimizing prediction errors.

Which method is better?

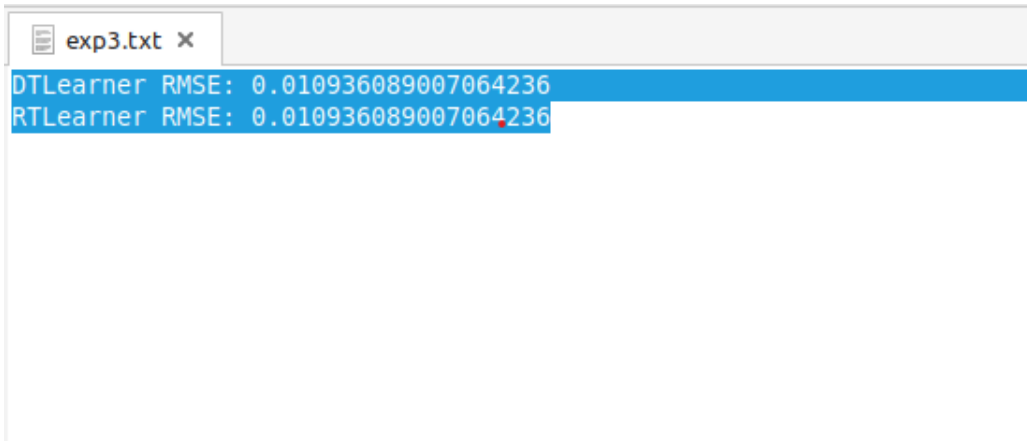
DTLearner had an RMSE of 0.0099, while RTLearner achieved a lower RMSE of 0.0083 thus RTLearner was better.

Which learner had better performance?

RTLearner performed much better, as it was more successful at minimizing RMSE compared than DTLearner.

Is one learner superior?

RTLearner performed better in this experiment, but DTLearner might still be preferred in other contexts for instance in the case when it is necessary to minimize larger outlier errors

A screenshot of a terminal window with a tab labeled 'exp3.txt'. The terminal displays two lines of text: 'DTLearner RMSE: 0.010936089007064236' and 'RTLearner RMSE: 0.010936089007064236'. The second line is highlighted in blue, and a red cursor is visible at the end of the text.

```
exp3.txt x
DTLearner RMSE: 0.010936089007064236
RTLearner RMSE: 0.010936089007064236
```

6 SUMMARY

In summary, this report explored overfitting in DTLearner and how bagging reduces its impact.

Experiment 1 (DTLearner): Contrary to expectations, no significant overfitting was observed in DTLearner.

Experiment 2 (Bagging with DTLearner): Bagging reduced the variance in RMSE and stabilized performance across leaf sizes.

Experiment 3 (Comparison of DTLearner and RTLearner): RTLearner outperformed DTLearner based on RMSE, with RTLearner achieving a lower RMSE (0.0083) compared to DTLearner (0.0099). Suggesting the RTLearner is more effective at minimizing predication error.

DTLearner and RTLearner have unique advantages.

In conclusion, while Bagging was effective in reducing overfitting, more complex datasets are needed to further explore the behavior of DTLearner in terms of overfitting. RTLearner proved to be superior in terms of RMSE however the best learner is context dependent based on prioritized metrics.

7 CHARTS

